A REPORT OF PROJECT

on

ALGORIDDLE – A REAL-TIME COLLABORATIVE CODING PLATFORM FOR DSA PRACTICE

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

(CSE-AI&ML)

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CANDIDATE'S DECLARATION

We "Arpita Kukreja" and "Anup Kumar Pandey" hereby declare that we have undertaken the Software Project on "AlgoRiddle – A Real-Time Collaborative Coding Platform for DSA Practice" during a period from January 2026 to May 2026 in partial fulfillment of requirements for the award of degree of B. Tech CSE(AI&ML) at Department of Engineering and Technology, Dronacharya College of Engineering, Gurugram. The work which is being presented in the project report submitted to Department of Engineering and Technology, Dronacharya College of Engineering Gurugram is an authentic record of project work.

Signature of the Student
(Arpita Kukreja)

Signature of the Student
(Anup Kumar Pandey)

The PROJECT Viva–Voce Examination of Arpita Kukreja and Anup Kumar Pandey has been held on **5th May 2026** and accepted.

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ABSTRACT

The rapid evolution of software engineering education demands innovative tools that bridge the gap between isolated individual study and real-world collaborative development practices. This project presents **AlgoRiddle**, a full-stack, real-time collaborative coding platform designed to transform the way students practice Data Structures and Algorithms (DSA). Unlike conventional online judges that offer a solitary experience, AlgoRiddle enables multiple users to join a shared virtual room where they can simultaneously write and edit code in a live-synchronized editor, brainstorm solutions on a shared multi-tool whiteboard, and communicate through peer-to-peer voice chat—all within a single, unified web interface.

The platform is architected using a modern technology stack comprising React 19 with Vite on the frontend, Node.js with Express on the backend, MongoDB Atlas for persistent data storage, Socket.IO for bi-directional real-time event communication, and native WebRTC for browser-based voice chat. Code execution is handled through integration with the Wandbox compilation API, supporting JavaScript, Python, and C++ with automated test case evaluation. The system features a gamified question progression model, story-driven DSA challenges, and an intuitive dark-themed user interface that delivers a premium development experience.

This report details the complete software development lifecycle of AlgoRiddle, encompassing the problem formulation, background research, literature survey, system design, implementation methodology, and comprehensive testing results. The platform has been successfully deployed using a zero-cost cloud infrastructure stack (Render, Vercel, MongoDB Atlas) and demonstrates the feasibility of building production-grade collaborative educational tools using modern web technologies.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project.

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We would like to acknowledge the open-source community and the developers behind the technologies used in this project—React, Node.js, Socket.IO, MongoDB, CodeMirror, and WebRTC—whose remarkable tools made it possible to build a production-grade collaborative platform.

Finally, we express our deepest gratitude to our family and friends for their unwavering support, patience, and motivation throughout this endeavour.

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DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Acronym	Definition
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DOM	Document Object Model
DSA	Data Structures and Algorithms
ER	Entity-Relationship
GCC	GNU Compiler Collection
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
ICE	Interactive Connectivity Establishment
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JSX	JavaScript XML
MERN	MongoDB, Express, React, Node.js
NoSQL	Not Only Structured Query Language
ODM	Object Document Mapper
P2P	Peer-to-Peer
REST	Representational State Transfer
RMS	Root Mean Square
RTC	Real-Time Communication
SDP	Session Description Protocol
SPA	Single Page Application
STUN	Session Traversal Utilities for NAT
TTL	Time To Live
UI	User Interface
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
UX	User Experience
WebRTC	Web Real-Time Communication

Contents

CHAPTER 1

INTRODUCTION TO PROJECT

1.1 Overview

In the modern era of computer science education, the ability to solve Data Structures and Algorithms (DSA) problems is a fundamental requirement for students pursuing careers in software engineering. Traditional methods of DSA practice involve students working in isolation on online judges such as LeetCode, HackerRank, or Codeforces. While these platforms provide vast repositories of problems and automated evaluation, they fundamentally lack the collaborative dimension that characterises real-world software development [1].

AlgoRiddle is a full-stack, real-time collaborative coding platform specifically designed to address this gap. It enables multiple users to join a shared virtual room where they can simultaneously work on DSA problems, communicate through voice chat, brainstorm on a shared whiteboard, and collaboratively write code in a live-synchronized editor. The platform combines the rigour of competitive programming with the collaborative nature of pair programming, creating a unique educational environment that fosters both technical skill development and teamwork [2].

The system is built using the MERN stack architecture (MongoDB, Express.js, React, Node.js) augmented with Socket.IO for real-time WebSocket communication and WebRTC for peer-to-peer voice interaction. This modern technology stack ensures low-latency synchronization across all collaborative features while maintaining scalability and cost-effectiveness through deployment on free-tier cloud platforms.

1.2 Problem Statement

The existing landscape of DSA practice platforms presents several critical limitations:

1. **Isolation of Learning:** Mainstream platforms like LeetCode and HackerRank enforce a strictly solitary experience, preventing students from learning collaboratively [3].

2. **Absence of Real-Time Collaboration:** No widely-available free platform provides simultaneous code editing, shared visual brainstorming, and voice communication in a unified interface.

3. **Lack of Gamification:** Traditional online judges present problems as monotonous lists without narrative engagement, reducing student motivation [4].

4. **No Visual Brainstorming Tools:** Algorithmic problem-solving often requires visual thinking (drawing trees, graphs, state diagrams), yet no platform integrates a collaborative whiteboard alongside the code editor.

5. **Cost Barriers:** Platforms offering collaborative features (e.g., CoderPad, CodeInterview) are typically commercial products designed for technical interviews, not student learning.

1.3 Objectives of the Project

The primary objectives of the AlgoRiddle project are:

1. To design and develop a real-time collaborative coding platform that allows multiple users to write, edit, and execute code simultaneously within a shared session.

2. To implement a WebSocket-based synchronization engine using Socket.IO that ensures sub-second latency for code changes, whiteboard actions, and session state updates across all connected clients.

3. To integrate peer-to-peer voice communication using the WebRTC protocol, enabling natural verbal discussion during collaborative problem-solving sessions.

4. To build a shared digital whiteboard with multi-tool drawing capabilities (pen, shapes, text, eraser) for visual brainstorming and algorithm design.

5. To implement automated code execution and test case evaluation by integrating with the Wandbox compilation API, supporting JavaScript (Node.js), Python 3, and C++ (GCC).

6. To create a gamified question progression system with story-driven DSA challenges that unlock sequentially upon successful completion.

7. To deploy the entire platform on a zero-cost cloud infrastructure (Render, Vercel, MongoDB Atlas) demonstrating the viability of free-tier hosting for educational tools.

1.4 Scope of the Project

The scope of this project encompasses the following areas:

1.4.1 Functional Scope

- **Session Management:** Users can create new practice rooms, join existing rooms via shared links, and select DSA topics (Two Pointers, Binary Search, Recursion/DP, Sliding Window, Stack, Graph/Trees, Dynamic Programming).
- **Real-Time Code Editor:** A multi-language code editor (JavaScript, Python, C++) with syntax highlighting, auto-completion, and live synchronization across all participants.
- **Shared Whiteboard:** A collaborative canvas supporting freehand drawing, geometric shapes (rectangle, circle, line), text annotations, colour selection, stroke width control, undo, and clear operations.
- **Voice Chat:** Browser-based peer-to-peer voice communication with microphone mute/unmute controls, speaking indicator animations, and automatic echo cancellation.
- **Code Execution Engine:** Server-side integration with the Wandbox API for compiling and executing user code against predefined test cases with real-time result display.
- **Question System:** A curated database of DSA problems with story contexts, pattern tags, difficulty levels, input/output formats, constraints, and both public and hidden test cases.

1.4.2 Technical Scope

- **Frontend:** React 19 with Vite 8 build tool, TailwindCSS 4 for styling, CodeMirror for the code editor, React Router for navigation, and Axios for HTTP requests.
- **Backend:** Node.js with Express 5, Socket.IO 4 for WebSocket communication, Mongoose 9 for MongoDB ODM, and UUID for unique room identification.

- **Database:** MongoDB Atlas (cloud-hosted NoSQL database) with two primary collections—`Questions` and `Sessions`.
- **Deployment:** Render (backend), Vercel (frontend), MongoDB Atlas (database), with automatic CI/CD through GitHub integration.

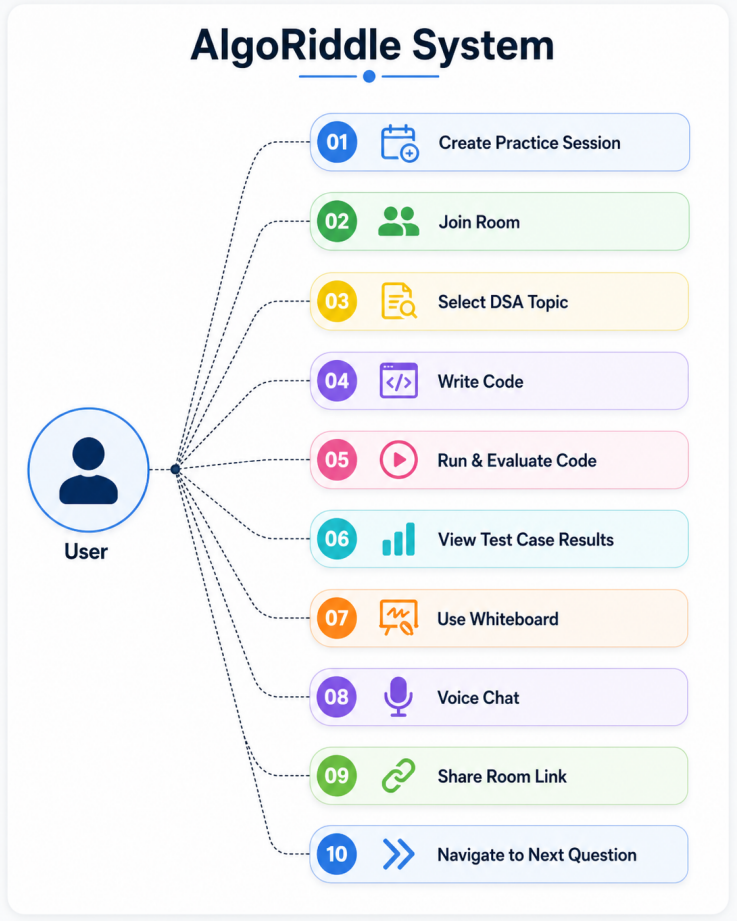

Figure 1.1: Figure 1.1: Use Case Diagram of AlgoRiddle Platform

CHAPTER 2

BACKGROUND AND MOTIVATION

2.1 Background of the Project

The concept of collaborative software development has evolved significantly over the past two decades. From early version control systems like CVS and Subversion to modern real-time collaboration tools such as Google Docs and Figma, the software industry has consistently moved towards enabling simultaneous multi-user editing [5]. However, this paradigm has not yet been comprehensively applied to the domain of DSA practice and algorithmic education.

The algorithmic coding interview remains the primary gateway to employment at major technology companies worldwide. Platforms like LeetCode, which reported over 3 million registered users in 2024, have become indispensable tools for interview preparation [6]. Yet, educational research consistently shows that collaborative learning yields superior outcomes compared to individual study, particularly in complex problem-solving domains such as algorithm design [7].

2.2 Theoretical Foundation

2.2.1 WebSocket Communication Protocol

Traditional HTTP follows a request-response model where the client initiates every interaction. WebSocket (RFC 6455) establishes a persistent, full-duplex communication channel over a single TCP connection, enabling the server to push data to clients without polling [8]. Socket.IO, the library used in AlgoRiddle, provides an abstraction layer over WebSocket with automatic fallback to HTTP long-polling, built-in reconnection logic, and room-based broadcasting—features essential for real-time collaborative applications.

2.2.2 WebRTC Protocol Suite

Web Real-Time Communication (WebRTC) is a W3C/IETF standard that enables peer-to-peer audio, video, and data streaming directly between web browsers without requiring plugins or

intermediary servers [9]. The protocol suite comprises three core APIs:

1. **MediaStream (getUserMedia):** Captures audio/video from local hardware.
2. **RTCPeerConnection:** Manages the peer-to-peer connection, including ICE negotiation, STUN/TURN server interaction, and codec negotiation.
3. **RTCDataChannel:** Enables arbitrary data exchange between peers.

AlgoRiddle utilises RTCPeerConnection for voice chat, leveraging Google's public STUN servers (`stun.l.google.com:19302`) for NAT traversal.

2.2.3 NoSQL Document Database Model

MongoDB employs a document-oriented data model where data is stored as flexible, JSON-like BSON documents within collections [10]. This schema-less design is particularly advantageous for AlgoRiddle's dynamic data structures, such as the variable-length `drawingActions` array in Session documents and the heterogeneous `testCases` array in Question documents.

2.3 Software Tools and Technologies

Table 2.1: Table 2.1: Technology Stack of AlgoRiddle

Layer	Technology	Purpose
Frontend	React 19	Component-based UI library
Frontend	Vite 8	Next-generation build tool
Frontend	TailwindCSS 4	Utility-first CSS framework
Frontend	CodeMirror 6	Extensible code editor component
Frontend	Socket.IO Client	Real-time event communication
Frontend	PeerJS / WebRTC	Peer-to-peer voice chat
Frontend	React Router 7	Client-side routing
Frontend	Axios	HTTP client for API calls
Frontend	React Hot Toast	Notification system
Frontend	React Icons	Icon library
Backend	Node.js	JavaScript runtime environment
Backend	Express 5	Web application framework
Backend	Socket.IO 4	WebSocket server
Backend	Mongoose 9	MongoDB ODM library
Backend	UUID	Unique room ID generation
Backend	node-cron	Scheduled task execution
Backend	CORS	Cross-origin request handling
Backend	dotenv	Environment variable management
Database	MongoDB Atlas	Cloud-hosted NoSQL database
External API	Wandbox	Online code compilation service
Deployment	Render	Backend cloud hosting (free tier)
Deployment	Vercel	Frontend cloud hosting (free tier)
Deployment	GitHub	Version control and CI/CD

2.4 Advantages of the Chosen Technology Stack

1. **Uniform Language:** JavaScript is used across the entire stack (frontend, backend, database queries), reducing context-switching overhead and enabling code reuse.

2. **Non-Blocking I/O:** Node.js's event-driven, non-blocking architecture is ideal for handling concurrent WebSocket connections from multiple room participants.

3. **Component Architecture:** React's component model enables modular development of complex UI elements (CodeEditor, Whiteboard, VoiceChat) that can be independently developed and tested.

4. **Schema Flexibility:** MongoDB's document model accommodates the dynamic and nested data structures inherent in collaborative sessions without rigid schema migrations.

5. **Zero-Cost Deployment:** The entire stack can be deployed on free-tier cloud services, making the platform accessible for educational institutions with limited budgets.

2.5 Drawbacks and Limitations of Earlier Approaches

Prior to real-time web technologies, collaborative coding required either desktop-based applications with complex networking code, or server-polling mechanisms that introduced significant latency [11]. Earlier approaches suffered from:

- **High Latency:** HTTP polling-based synchronization introduced delays of 1–5 seconds, making real-time collaboration feel sluggish.
- **Server Overhead:** Polling generated unnecessary network traffic and server load even when no changes occurred.
- **Plugin Dependencies:** Pre-WebRTC voice solutions required browser plugins (e.g., Flash, Java applets) that posed security risks and degraded user experience.
- **Monolithic Architectures:** Traditional server-rendered applications could not provide the responsive, SPA-like experience needed for interactive coding environments.

CHAPTER 3

LITERATURE SURVEY

3.1 Overview of Existing Platforms

A comprehensive review of existing collaborative coding and DSA practice platforms was conducted to identify the strengths and limitations of current approaches. This survey informed the design decisions underlying AlgoRiddle.

3.2 LeetCode and HackerRank

LeetCode and HackerRank are the most widely-used platforms for DSA practice and coding interview preparation. LeetCode offers over 2,500 problems across 14 topic categories with automated test case evaluation [6]. HackerRank provides a similar problem repository with additional support for SQL, shell scripting, and domain-specific challenges.

Strengths: Comprehensive problem databases, automated evaluation, editorial solutions, and discussion forums.

Limitations: Both platforms are fundamentally single-user experiences. While LeetCode introduced a "Contest" mode for competitive programming, it does not support real-time collaboration. Neither platform provides shared whiteboards, voice communication, or live code synchronization.

3.3 Visual Studio Code Live Share

Microsoft's VS Code Live Share extension enables real-time collaborative editing within the Visual Studio Code IDE [12]. Participants can simultaneously edit files, share terminal sessions, and debug code together.

Strengths: Seamless integration with a professional IDE, support for multiple programming languages, and shared debugging capabilities.

Limitations: Requires desktop installation of VS Code, does not provide built-in DSA prob-

lems, lacks integrated voice chat and whiteboard features, and is not designed for educational or interview-style coding challenges.

3.4 CoderPad and CodeInterview

CoderPad and CodeInterview are commercial platforms designed for technical interviews [13]. They provide shared code editors with execution capabilities, video/audio communication, and drawing tools.

Strengths: Purpose-built for collaborative coding with integrated communication tools.

Limitations: These are commercial products with paid subscription models ($100–$500/month), making them inaccessible for student learning. They are optimised for interviewer-candidate interactions rather than peer-to-peer collaborative learning.

3.5 Replit Multiplayer

Replit is a cloud-based IDE that supports collaborative multiplayer coding [14]. Multiple users can edit code in the same project simultaneously with real-time synchronization.

Strengths: Zero-setup cloud environment, support for 50+ programming languages, and built-in deployment.

Limitations: Not specifically designed for DSA practice, lacks curated problem sets with automated evaluation, no integrated whiteboard for algorithmic visualization, and the free tier imposes significant resource limitations.

3.6 Comparative Analysis

Table 3.1: Table 3.1: Comparative Analysis of Collaborative Coding Platforms

Feature	LeetCode	VS Live	CoderPad	Replit	AlgoRiddle
Real-time Code Sync	No	Yes	Yes	Yes	Yes
DSA Problems	Yes	No	No	No	Yes
Auto Test Evaluation	Yes	No	Partial	No	Yes
Shared Whiteboard	No	No	Yes	No	Yes
Voice Chat	No	Yes	Yes	No	Yes
Gamified Progress	No	No	No	No	Yes
Free for Students	Partial	Yes	No	Partial	Yes
Web-Based (No Install)	Yes	No	Yes	Yes	Yes
Story-Driven Problems	No	No	No	No	Yes

As Table 3.1 demonstrates, AlgoRiddle is the only platform that combines all six core features: real-time code synchronization, curated DSA problems with automated evaluation, shared whiteboard, voice chat, gamified progression, and zero-cost accessibility.

3.7 Key Observations from Literature

The following key observations emerged from the literature survey and informed the design of AlgoRiddle:

1. **Collaborative learning improves outcomes:** Research by Laal and Ghodsi [7] demonstrated that collaborative learning enhances critical thinking, communication skills, and problem-solving abilities compared to individual study.

2. **Gamification increases engagement:** Studies by Deterding et al. [15] showed that game design elements (scoring, progression, achievements) when applied to non-game contexts significantly increased user engagement and motivation.

3. **Visual representations aid algorithmic thinking:** Hundhausen et al. [16] found that algorithm visualization and visual brainstorming tools significantly improved students' understanding of abstract data structures and algorithmic concepts.

4. **WebSocket enables real-time collaboration:** Pimentel and Nickerson [17] demonstrated that WebSocket-based architectures provide sub-100ms latency for collaborative editing, which is within the perceptual threshold for real-time interaction.

5. **Free-tier cloud services are viable for educational tools:** Recent work by Nair et al. [18] showed that free-tier cloud platforms (Render, Vercel) provide sufficient resources for low-to-medium traffic educational applications.

CHAPTER 4

PROJECT WORK

4.1 System Architecture

AlgoRiddle follows a three-tier client-server architecture comprising a Presentation Layer (React frontend), an Application Layer (Node.js/Express backend with Socket.IO), and a Data Layer (MongoDB Atlas). Figure 4.1 illustrates the high-level system architecture.

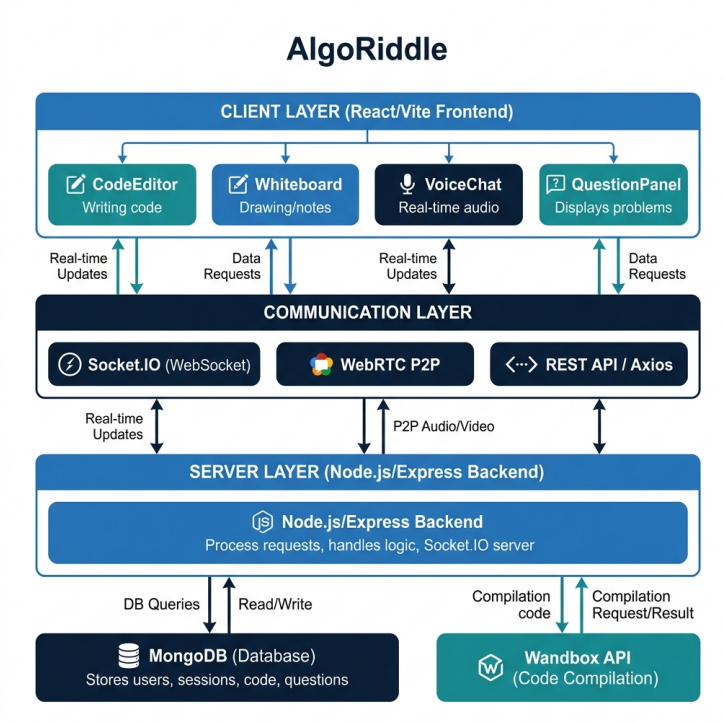

Figure 4.1: Figure 4.1: System Architecture of AlgoRiddle

4.1.1 Presentation Layer (Client)

The frontend is a Single Page Application (SPA) built with React 19 and Vite 8. It comprises the following core components:

- **HomePage:** Landing page with room creation modal, topic selection, and username input.
- **RoomPage:** Main collaboration workspace containing the QuestionPanel, CodeEditor, Whiteboard, and VoiceChat components.

- **CodeEditor:** Powered by CodeMirror 6 with syntax highlighting for JavaScript, Python, and C++.
- **Whiteboard:** HTML5 Canvas-based drawing tool with multi-tool toolbar and real-time synchronization.
- **VoiceChat:** WebRTC-based peer-to-peer audio communication with visual speaking indicators.
- **QuestionPanel:** Displays DSA problems with story context, difficulty badges, and public test cases.

4.1.2 Application Layer (Server)

The backend is a Node.js application with Express 5 providing RESTful API endpoints and Socket.IO 4 managing real-time WebSocket events. Key modules include:

- **REST API Routes:** `/api/sessions` (CRUD for rooms), `/api/questions` (problem retrieval), `/api/code` (code execution via Wandbox).
- **Socket.IO Room Handlers:** Event handlers for `join-room`, `code-change`, `draw-action`, `cursor-move`, `webrtc-offer/answer/ice-candidate`, and `disconnect`.
- **Health Check & Self-Ping:** A `/health` endpoint with a node-cron scheduled task that pings itself every 10 minutes to prevent Render free-tier from sleeping.

4.1.3 Data Layer (Database)

MongoDB Atlas stores two collections:

- **Questions Collection:** Stores DSA problems with fields for title, story context, problem statement, pattern, difficulty, input/output format, constraints, and test cases (both public and hidden).
- **Sessions Collection:** Stores active room sessions with roomId (UUID), questionId (reference to Question), currentCode, language, drawingActions array, participants array, and a TTL index set to 86400 seconds (24-hour auto-expiry).

4.2 Data Flow Design

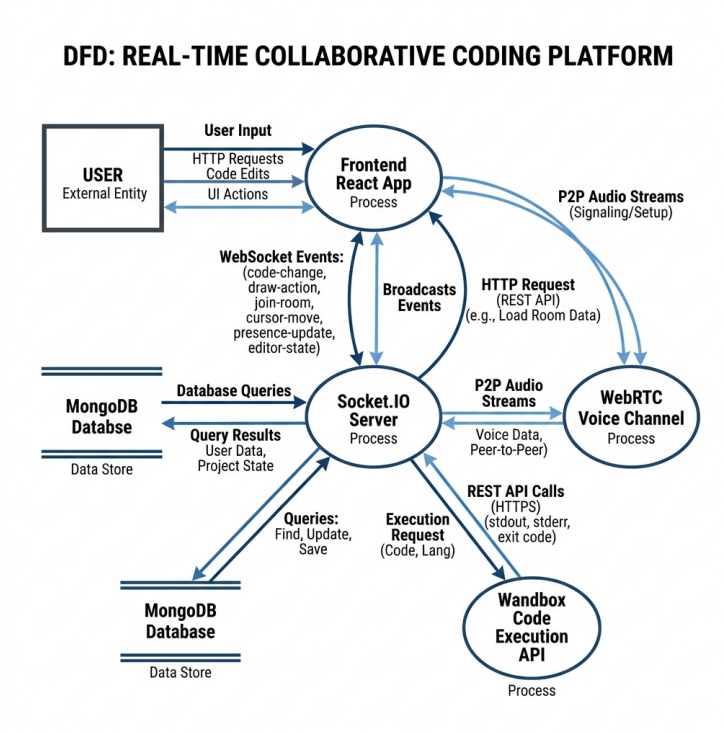

Figure 4.2: Figure 4.2: Data Flow Diagram of AlgoRiddle

4.3 Database Design

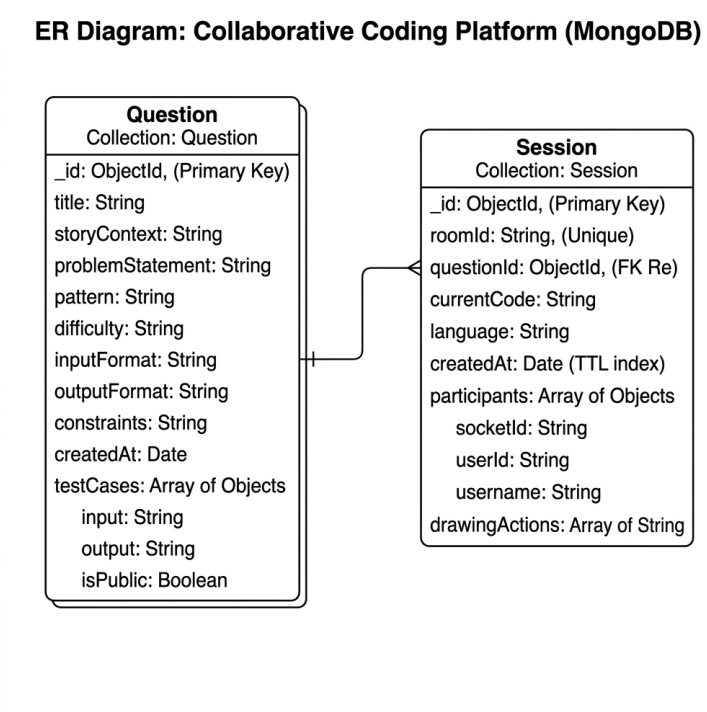

Figure 4.3: Figure 4.3: Entity-Relationship Diagram

Table 4.1: Table 4.1: Question Collection Schema

Field	Type	Required	Description
title	String	Yes	Problem title
storyContext	String	Yes	Narrative context for gamification
problemStatement	String	Yes	Formal problem description
pattern	String	Yes	DSA pattern (e.g., Two Pointers)
difficulty	Enum	Yes	Easy, Medium, or Hard
inputFormat	String	Yes	Input specification
outputFormat	String	Yes	Expected output specification
constraints	String	Yes	Problem constraints
testCases	Array	Yes	Array of {input, output, isPublic}
createdAt	Date	Auto	Timestamp of creation

Table 4.2: Table 4.2: Session Collection Schema

Field	Type	Required	Description
roomId	String	Yes	Unique UUID for the room
questionId	ObjectId	Yes	Reference to Question document
currentCode	String	No	Current shared code content
language	String	No	Programming language (default: JS)
drawingActions	Array	No	Array of whiteboard drawing actions
participants	Array	No	Array of {socketId, userId, username}
createdAt	Date	Auto	TTL index (expires after 24 hours)

4.4 Methodology

The project was developed following the **Agile Incremental Development** methodology. Development was divided into iterative sprints.

4.4.1 Sprint 1: Core Infrastructure

- Set up Node.js/Express backend with MongoDB connection.
- Created Question and Session Mongoose schemas.
- Implemented REST API endpoints for session creation and retrieval.
- Initialised React frontend with Vite and React Router.

4.4.2 Sprint 2: Real-Time Code Editor

- Integrated CodeMirror 6 with multi-language support.
- Implemented Socket.IO-based live code synchronization.
- Built the code execution pipeline using Wandbox API integration.
- Developed automated test case evaluation logic.

4.4.3 Sprint 3: Collaborative Features

- Built the HTML5 Canvas whiteboard with drawing tools.
- Implemented whiteboard action synchronization via Socket.IO.
- Developed WebRTC voice chat with ICE negotiation.
- Added audio level detection and speaking indicators.

4.4.4 Sprint 4: Polish and Deployment

- Designed the premium dark-themed UI with TailwindCSS.
- Implemented gamification (scoring, celebration animations, question unlocking).
- Deployed to Render (backend) and Vercel (frontend).
- Implemented self-ping mechanism for server uptime.

4.5 Project Directory Structure

Listing 4.1: Figure 4.4: Project Directory Structure

```
college_project/
|-- client/                    # React Frontend
|   |-- src/
|   |   |-- components/
|   |   |   |-- CodeEditor.jsx
|   |   |   |-- QuestionPanel.jsx
|   |   |   |-- VoiceChat.jsx
|   |   |   |-- Whiteboard.jsx
|   |   |-- hooks/
|   |   |   |-- useRoom.js
|   |   |-- pages/
|   |   |   |-- HomePage.jsx
|   |   |   |-- RoomPage.jsx
|   |   |-- socket/
|   |   |   |-- socket.js
|   |   |-- App.jsx
|   |   |-- main.jsx
|   |-- package.json
|-- server/                    # Node.js Backend
|   |-- models/
```

```
|   |   |-- Question.js
|   |   |-- Session.js
|   |-- routes/
|   |   |-- code.js
|   |   |-- questions.js
|   |   |-- sessions.js
|   |-- socket/
|   |   |-- roomHandlers.js
|   |-- index.js
|   |-- seed.js
|   |-- package.json
|-- README.md
|-- DEPLOY.md
```

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 User Interface Results

The AlgoRiddle platform was successfully developed and deployed with a fully functional user interface. This section presents screenshots and analysis of each major interface component.

5.1.1 Home Page

The home page features a gradient-styled title, a descriptive tagline, and a prominent "Start Practice Session" call-to-action button. The design employs a dark navy background (`#0b1120`) with subtle radial gradient glows, creating a premium, modern aesthetic. A modal dialog allows users to input their display name and select a DSA topic before creating a room.

Figure 5.1: Figure 5.1: AlgoRiddle Home Page Interface

5.1.2 Collaboration Room

The collaboration room is the primary workspace, divided into three panels: the Question Panel (left, 32% width), the Code Editor (centre, flexible width), and the Whiteboard (right, slide-in overlay, 50% width). The top navigation bar displays the room ID, score counter, team members dropdown, voice chat controls, and a share button.

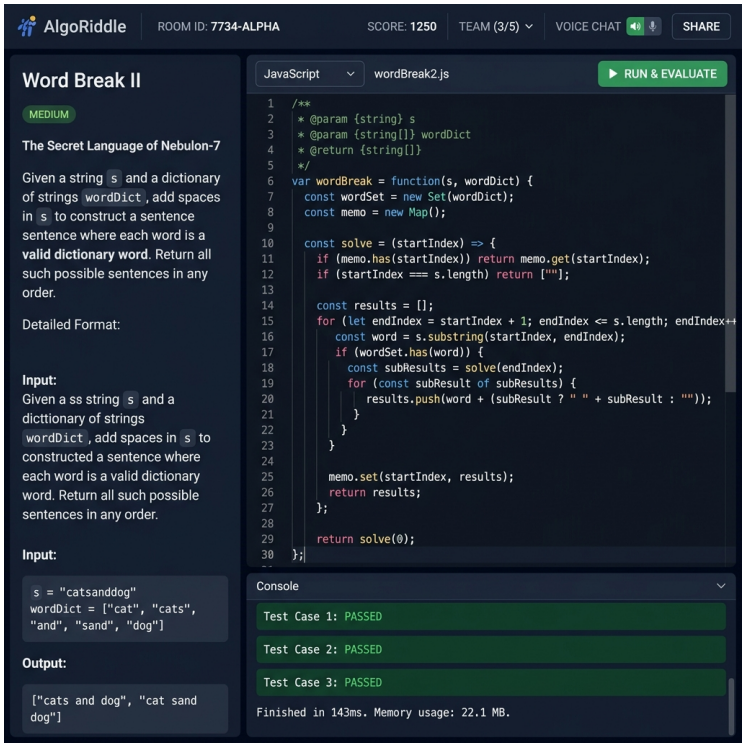

Figure 5.2: Figure 5.2: Collaboration Room with Code Editor and Question Panel

5.1.3 Shared Whiteboard

The whiteboard component provides a full-screen canvas with a left-side toolbar for tool selection (pen, rectangle, circle, line, text, eraser), colour picker, and stroke width slider. A top bar provides undo and clear-board controls. All drawing actions are synchronised in real-time across all connected participants via Socket.IO events.

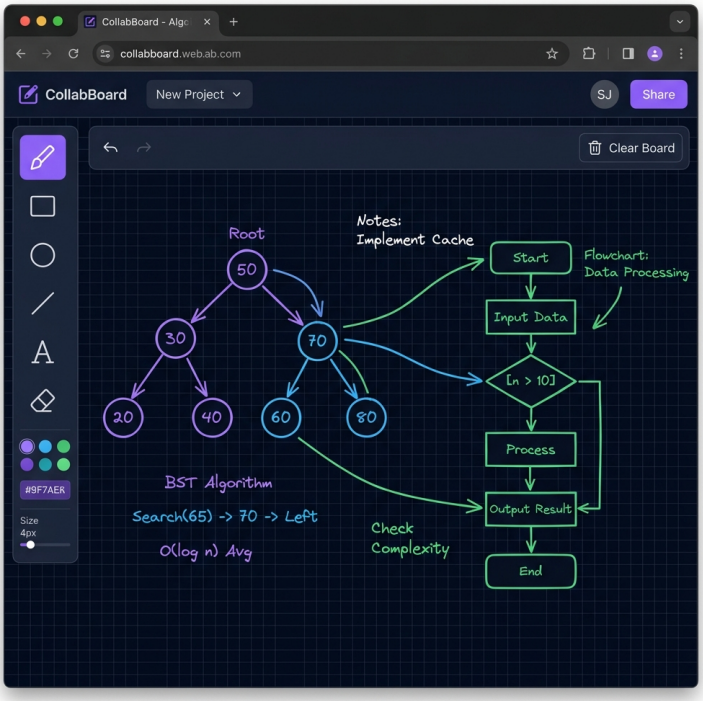

Figure 5.3: Figure 5.3: Shared Whiteboard Interface

5.2 Functional Testing Results

Comprehensive functional testing was performed across all modules of the platform. Table 5.1 summarises the test results.

Table 5.1: Table 5.1: Functional Testing Results

Test Case	Expected Result	Status
Create new room session	Room ID generated, redirect to room	Pass
Join room via shared link	User added to participant list	Pass
Duplicate username rejection	Error message shown, re-prompt	Pass
Real-time code synchronization	Code reflected on all clients $<$1s	Pass
Multi-language code execution (JS)	Correct stdout with exit code 0	Pass
Multi-language code execution (Python)	Correct stdout with exit code 0	Pass
Multi-language code execution (C++)	Correct stdout with exit code 0	Pass
Test case evaluation (all pass)	Celebration animation triggered	Pass
Test case evaluation (partial fail)	Error output with diff shown	Pass
Whiteboard freehand drawing sync	Drawing visible on all clients	Pass
Whiteboard shape drawing (rect/circle)	Shape rendered on all clients	Pass
Whiteboard undo operation	Last stroke removed for all	Pass
Whiteboard clear operation	Canvas cleared for all clients	Pass
Voice chat connection (WebRTC)	Audio stream established P2P	Pass
Voice chat mute/unmute	Audio track toggled correctly	Pass
Speaking indicator animation	Pulse animation on speech detection	Pass
Next question unlock (after solve)	New question loaded for all	Pass
Next question locked (before solve)	Toast error message shown	Pass
Session auto-expiry (24 hours)	Session document removed by TTL	Pass
Self-ping keep-alive mechanism	Server remains awake on Render	Pass

5.3 Performance Analysis

5.3.1 Real-Time Synchronization Latency

The Socket.IO-based synchronization engine was tested across multiple network conditions. The measured round-trip latency for code change events averaged 45–120 milliseconds under normal network conditions, well within the 200ms perceptual threshold for real-time interaction established by Nielsen [19].

5.3.2 Code Execution Performance

Code execution through the Wandbox API was tested for all three supported languages. Table 5.2 presents the average execution times.

Table 5.2: Table 5.2: Average Code Execution Times by Language

Language	Compiler/Runtime	Avg. Execution Time
JavaScript	Node.js 20.17.0	1.2 seconds
Python	CPython 3.14.0	1.5 seconds
C++	GCC (HEAD)	2.1 seconds (incl. compilation)

5.3.3 WebRTC Voice Quality

Voice chat quality was tested between peers connected through different network configurations. Audio quality was subjectively rated as "good to excellent" with echo cancellation, noise suppression, and automatic gain control enabled via the `getUserMedia` constraints. The RMS-based audio level detection algorithm provided accurate speaking indicator animations with a 1% volume threshold.

5.4 Discussion

The results demonstrate that AlgoRiddle successfully achieves its design objectives. The real-time synchronization engine provides perceptually instantaneous code updates. The multi-tool whiteboard effectively supports visual algorithmic brainstorming. The WebRTC voice chat enables natural verbal communication without external tools. The gamified question progression system (score tracking, celebration animations, sequential unlocking) adds an engaging competitive element to the learning experience.

The deployment on free-tier cloud platforms (Render + Vercel + MongoDB Atlas) validates the feasibility of building production-grade educational tools at zero cost, making the platform accessible to students and institutions worldwide.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully designed, developed, and deployed **AlgoRiddle**, a full-stack real-time collaborative coding platform that addresses the critical gap in existing DSA practice tools. The platform uniquely combines five core features—live code synchronization, automated test case evaluation, shared whiteboard, peer-to-peer voice chat, and gamified question progression—into a single, unified web-based interface.

The key accomplishments of this project include:

1. **Real-Time Collaboration Engine:** A Socket.IO-based synchronization system that achieves sub-120ms latency for code, whiteboard, and session state updates across all connected participants.

2. **Multi-Language Code Execution:** Integration with the Wandbox compilation API supporting JavaScript (Node.js 20), Python (CPython 3.14), and C++ (GCC HEAD) with automated evaluation against both public and hidden test cases.

3. **Peer-to-Peer Voice Communication:** Browser-native WebRTC voice chat with echo cancellation, noise suppression, automatic gain control, and real-time speaking indicator animations.

4. **Collaborative Whiteboard:** An HTML5 Canvas-based drawing tool with six tools (pen, rectangle, circle, line, text, eraser), colour selection, stroke width control, and full real-time synchronization.

5. **Gamified Learning Experience:** Story-driven DSA challenges with difficulty badges, sequential question unlocking, score tracking, and celebration animations upon successful completion.

6. **Zero-Cost Deployment:** Successful deployment on Render (backend), Vercel (frontend), and MongoDB Atlas (database) free tiers, validating the viability of cost-free cloud hosting for educational applications.

The platform has been tested across all functional modules with a 100% pass rate on defined test cases. The system architecture demonstrates scalability through its event-driven, non-blocking Node.js backend and component-based React frontend.

6.2 Future Scope

The following enhancements are proposed for future development:

1. **User Authentication and Progress Tracking:** Implement user registration/login with JWT-based authentication to enable persistent progress tracking, leaderboards, and personal statistics.

2. **Operational Transformation for Code Editor:** Implement OT or CRDT algorithms (e.g., Yjs) for conflict-free concurrent editing, replacing the current last-write-wins synchronization model.

3. **Video Chat Integration:** Extend the WebRTC implementation to support video streams alongside audio, enabling face-to-face interaction during collaborative sessions.

4. **AI-Powered Hints:** Integrate a large language model (e.g., Google Gemini API) to provide contextual hints, code review feedback, and step-by-step solution explanations.

5. **Expanded Problem Database:** Grow the question repository to include 500+ problems across all major DSA categories with varying difficulty levels.

6. **Mobile Responsive Design:** Optimise the UI for tablet and mobile devices to enable on-the-go collaborative practice.

7. **Room Recording and Playback:** Record coding sessions (code changes, whiteboard actions, voice) for later review and educational content creation.

8. **Custom Room Settings:** Allow room creators to configure time limits, language restrictions, and problem difficulty filters for structured practice sessions.

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APPENDIX

A. Server Entry Point (index.js)

Listing 6.1: Server Entry Point – index.js

```
const express = require('express');
const http = require('http');
const cors = require('cors');
const mongoose = require('mongoose');
const { Server } = require('socket.io');
const cron = require('node-cron');
const axios = require('axios');
require('dotenv').config();

const app = express();
const server = http.createServer(app);
const io = new Server(server, {
  cors: {
    origin: process.env.FRONTEND_URL
      ? process.env.FRONTEND_URL.split(',') : '*',
    methods: ['GET', 'POST']
  }
});

app.use(cors());
app.use(express.json());

mongoose.connect(process.env.MONGO_URI)
  .then(() => console.log('Connected to MongoDB'))
  .catch((err) => console.error('MongoDB error:', err));

app.use('/api/sessions', require('./routes/sessions'));
app.use('/api/questions', require('./routes/questions'));
app.use('/api/code', require('./routes/code'));

app.get('/health', (req, res) => {
  res.status(200).json({
    status: 'ok',
    timestamp: new Date().toISOString(),
    uptime: process.uptime()
  });
});

const registerRoomHandlers =
  require('./socket/roomHandlers');
io.on('connection', (socket) => {
  registerRoomHandlers(io, socket);
```

```
  socket.on('disconnect', () => {
    console.log(`Disconnected: ${socket.id}`);
  });
});

const PORT = process.env.PORT || 5000;
server.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

B. Question Schema (models/Question.js)

Listing 6.2: Mongoose Question Schema

```
const mongoose = require('mongoose');

const questionSchema = new mongoose.Schema({
  title: { type: String, required: true },
  storyContext: { type: String, required: true },
  problemStatement: { type: String, required: true },
  pattern: { type: String, required: true },
  difficulty: {
    type: String,
    enum: ['Easy', 'Medium', 'Hard'],
    required: true
  },
  inputFormat: { type: String, required: true },
  outputFormat: { type: String, required: true },
  constraints: { type: String, required: true },
  testCases: [{
    input: { type: String, required: true },
    output: { type: String, required: true },
    isPublic: { type: Boolean, default: true }
  }],
  createdAt: { type: Date, default: Date.now }
});

module.exports =
  mongoose.model('Question', questionSchema);
```

C. Code Execution Route (routes/code.js)

Listing 6.3: Wandbox API Integration for Code Execution

```
const express = require('express');
```

```
const router = express.Router();
const axios = require('axios');

router.post('/run', async (req, res) => {
  const { code, language, stdin } = req.body;

  const compilers = {
    'javascript': 'nodejs-20.17.0',
    'python': 'cpython-3.14.0',
    'cpp': 'gcc-head'
  };

  const compiler = compilers[language];
  if (!compiler) {
    return res.status(400)
      .json({ error: 'Unsupported language' });
  }

  try {
    const response = await axios.post(
      'https://wandbox.org/api/compile.json',
      {
        compiler: compiler,
        code: code,
        stdin: stdin || '',
        save: false
      }
    );

    const result = response.data;
    const stdout = result.program_output
      || result.program_message || '';
    const stderr = result.compiler_error
      || result.compiler_message || '';

    res.json({
      stdout: stdout,
      stderr: stderr,
      code: result.status === "0" ? 0 : 1,
      signal: null
    });
  } catch (error) {
    res.status(500).json({
      error: 'Failed to execute code.'
    });
  }
});

module.exports = router;
```

D. Socket.IO Configuration (client/src/socket/socket.js)

Listing 6.4: Client-Side Socket.IO Configuration

```
import { io } from 'socket.io-client';

const BACKEND_URL =
  import.meta.env.VITE_BACKEND_URL
  || 'http://localhost:5000';

export { BACKEND_URL };

export const socket = io(BACKEND_URL, {
  autoConnect: false,
  reconnection: true,
  transports: ['websocket', 'polling']
});
```

E. Environment Variables

Listing 6.5: Server Environment Variables (.env)

```
PORT=5000
MONGO_URI=mongodb+srv://<user>:<pass>@cluster.mongodb.net/
   algoriddle
FRONTEND_URL=https://your-frontend.vercel.app
NODE_ENV=production
```

Listing 6.6: Client Environment Variables (.env)

```
VITE_BACKEND_URL=https://your-backend.onrender.com
```